

ICC-Tutorial: Worksheet for Safety Verification of Neural Networks using Starsets

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December 18, 2025

Goal

- Represent the input set as a star set.
- Apply affine transformation on a star set.
- Apply ReLU on a star set.
- Obtain the output star set(s).

Star Sets: Basic Facts

- A star set represents a set of points using a center, basis vectors, and linear constraints.
- A star set is written as $\theta = (c, V, P)$, where:
 - c is the center point,
 - V is the set of basis vectors,
 - P is a set of linear constraints, i.e., predicate of the form $A\alpha \leq B$.
- Each point in the star set has the form

$$x = c + V\alpha, \quad \text{where } \alpha \text{ satisfies } P.$$

- The predicate P restricts the range of the free variables α and defines the shape of the set.
- Affine transformations of a star set are exact: applying (W, b) to (c, V, P) gives $(Wc + b, WV, P)$.
- ReLU is handled by intersecting the star set with inequalities like $x_i \geq 0$ or $x_i < 0$, and updating the predicate accordingly. When a coordinate can be both positive and negative, ReLU splits the star set into multiple star sets.

Worksheet: Star Sets Through a ReLU Network

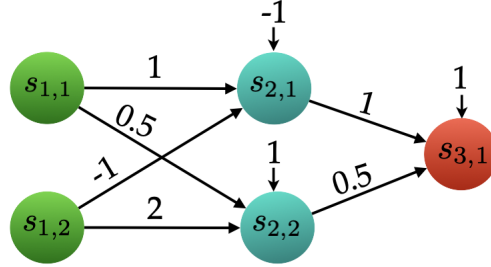


Figure 1: Neural Network $\mathcal{N}_1$

Use the network $\mathcal{N}$ shown in Fig. 1 (2 inputs, 2 hidden, 1 output). The network is $\mathcal{N} = \{(W_1, b_1), (W_2, b_2)\}$ with

$$W_1 = \begin{bmatrix} 1 & -1 \\ 0.5 & 2 \end{bmatrix}, \quad b_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \quad W_2 = \begin{bmatrix} 1 & -0.5 \end{bmatrix}, \quad b_2 = 1.$$

Input Star Set

Input star set: $(c_1, \{v_1, v_2\}, P_1)$:

$$c_1 = \begin{bmatrix} 1.5 \\ 1.5 \end{bmatrix}, \quad v_1 = \begin{bmatrix} 0.5 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 0 \\ 0.5 \end{bmatrix},$$

$$P_1 : \quad -1 \leq \alpha_1 \leq 1, \quad -1 \leq \alpha_2 \leq 1.$$

The star set is illustrated in Figure 2.

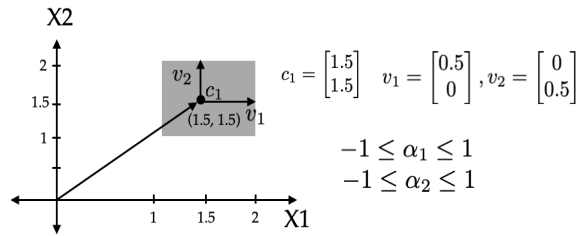


Figure 2: Input Star set

Verifying the geometry of input star set. Each point in the input star set has the form

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = c_1 + \alpha_1 v_1 + \alpha_2 v_2 = \begin{bmatrix} 1.5 \\ 1.5 \end{bmatrix} + \alpha_1 \begin{bmatrix} 0.5 \\ 0 \end{bmatrix} + \alpha_2 \begin{bmatrix} 0 \\ 0.5 \end{bmatrix} = \begin{bmatrix} 1.5 + 0.5\alpha_1 \\ 1.5 + 0.5\alpha_2 \end{bmatrix}.$$

Since $\alpha_1, \alpha_2 \in [-1, 1]$, we have

$$1.5 - 0.5 \leq x_1 \leq 1.5 + 0.5, \quad 1.5 - 0.5 \leq x_2 \leq 1.5 + 0.5.$$

That is,

$$x_1 \in [1, 2], \quad x_2 \in [1, 2].$$

Step 1: Affine transformation of the input star set

We apply the affine map (W_1, b_1) to the input star set

$$(c_1, \{v_1, v_2\}, P_1).$$

The affine transformation acts as follows:

$$\boxed{(c_1, \{v_1, v_2\}, P_1) \xrightarrow{(W_1, b_1)} (W_1 c_1 + b_1, \{W_1 v_1, W_1 v_2\}, P_1)}$$

The center and the basis vectors get transformed and the predicate P_1 remains unchanged. The resulting star set is denoted by

$$(c_2, \{v_3, v_4\}, P_1) \quad \text{where} \quad c_2 = W_1 c_1 + b_1, \quad v_3 = W_1 v_1, \quad v_4 = W_1 v_2$$

Compute the new center:

$$c_2 = W_1 c_1 = \begin{bmatrix} 1 & -1 \\ 0.5 & 2 \end{bmatrix} \begin{bmatrix} 1.5 \\ 1.5 \end{bmatrix} + \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 4.75 \end{bmatrix}.$$

Compute the new basis vectors:

$$v_3 = W_1 v_1 = \begin{bmatrix} 1 & -1 \\ 0.5 & 2 \end{bmatrix} \begin{bmatrix} 0.5 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.5 \\ 0.25 \end{bmatrix}, \quad v_4 = W_1 v_2 = \begin{bmatrix} 1 & -1 \\ 0.5 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 0.5 \end{bmatrix} = \begin{bmatrix} -0.5 \\ 1 \end{bmatrix}.$$

Step 2: ReLU at the hidden layer

We now apply the ReLU activation to the star set obtained after the first affine transformation. Recall that ReLU acts as $x \mapsto \max(0, x)$.

Since ReLU is applied element-wise, we handle one coordinate at a time. For each coordinate, we examine the star set and separate the points where the coordinate is nonnegative from those where it is negative. Points with nonnegative values are kept unchanged, while points with negative values are mapped to zero in that coordinate. This procedure may split the star set into multiple star sets, which we compute explicitly below.

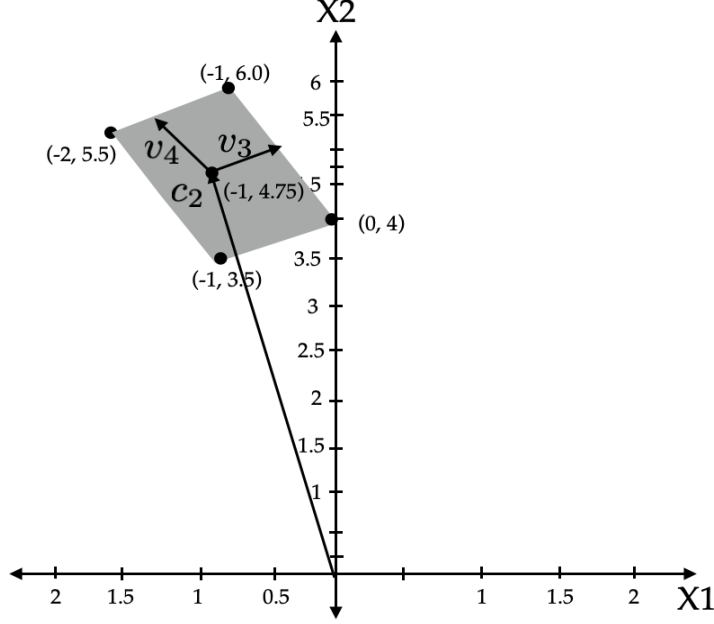


Figure 3: Star set after affine transformation at the hidden layer

Step 2(a): Applying ReLU to x_1 . We start from the star set

$$(c_2, \{v_3, v_4\}, P_1).$$

Step: Intersection with $x_1 \geq 0$.

$$(c_2, \{v_3, v_4\}, P_1) \cap (x_1 \geq 0) \longrightarrow (c_2, \{v_3, v_4\}, P_2)$$

That is, center and basis vectors remain same and P_2 captures

$$P_2 = P_1 \cap (x_1 \geq 0).$$

Through star set representation x_1 is written as

$$x_1 = -1 + 0.5\alpha_1 - 0.5\alpha_2.$$

Thus, the constraint $x_1 \geq 0$ becomes

$$-1 + 0.5\alpha_1 - 0.5\alpha_2 \geq 0 \iff -0.5\alpha_1 + 0.5\alpha_2 \leq -1.$$

Hence, the new predicate is

$$P_2 : \begin{cases} -1 \leq \alpha_1 \leq 1, \\ -1 \leq \alpha_2 \leq 1, \\ -0.5\alpha_1 + 0.5\alpha_2 \leq -1. \end{cases}$$

The only solution to these constraints is

$$\alpha_1 = 1 \quad \text{and} \quad \alpha_2 = -1$$

Hence, this restricts the star set to just one point as shown in Figure 4

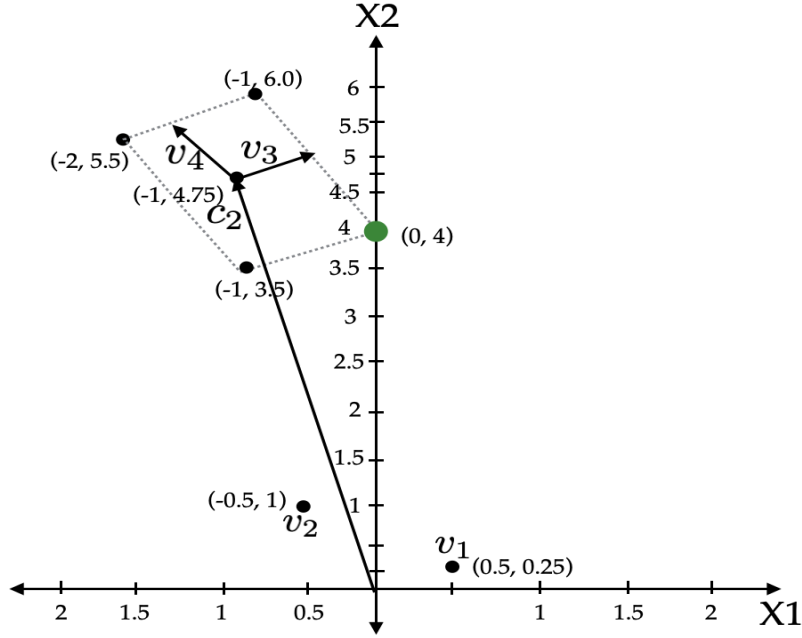


Figure 4: Star set after intersecting with $x_1 \geq 0$

Step: Intersection with $x_1 \leq 0$.

$$(c_2, \{v_3, v_4\}, P_1) \cap (x_1 \leq 0) \longrightarrow (c_2, \{v_3, v_4\}, P_3)$$

That is, the center and basis vectors remain the same and the predicate is updated to

$$P_3 = P_1 \cap (x_1 \leq 0).$$

Using the star set representation,

$$x_1 = -1 + 0.5\alpha_1 - 0.5\alpha_2.$$

Thus, the constraint $x_1 \leq 0$ becomes

$$-1 + 0.5\alpha_1 - 0.5\alpha_2 \leq 0 \iff 0.5\alpha_1 - 0.5\alpha_2 \leq 1.$$

Hence, the new predicate is

$$P_3 : \begin{cases} -1 \leq \alpha_1 \leq 1, \\ -1 \leq \alpha_2 \leq 1, \\ 0.5\alpha_1 - 0.5\alpha_2 \leq 1. \end{cases}$$

This set of constraints is non-empty, for example, $\alpha_1 = 0$ and $\alpha_2 = 0$ satisfies all constraints in P_3 .

ReLU update for the branch $x_1 \leq 0$.

To make $x_1 = 0$ of all the elements in the set, we do linear transformation with the following matrix.

$$T_1 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

We apply this linear map to the star set

$$(c_2, \{v_3, v_4\}, P_3).$$

The transformation acts as

$$(c_2, \{v_3, v_4\}, P_3) \xrightarrow{T_1} (T_1 c_2, \{T_1 v_3, T_1 v_4\}, P_3)$$

The resulting star set is denoted by $(c_3, \{v_5, v_6\}, P_3)$, where

$$c_3 = T_1 c_2, \quad v_5 = T_1 v_3, \quad v_6 = T_1 v_4.$$

Thus,

$$c_3 = T_1 c_2 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 4.75 \end{bmatrix} = \begin{bmatrix} 0 \\ 4.75 \end{bmatrix},$$

$$v_5 = T_1 v_3 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0.5 \\ 0.25 \end{bmatrix} = \begin{bmatrix} 0 \\ 0.25 \end{bmatrix}, \quad v_6 = T_1 v_4 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -0.5 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Hence, this restricts the star set to a straight line representing the projection-illustrated by the red line in Figure 5

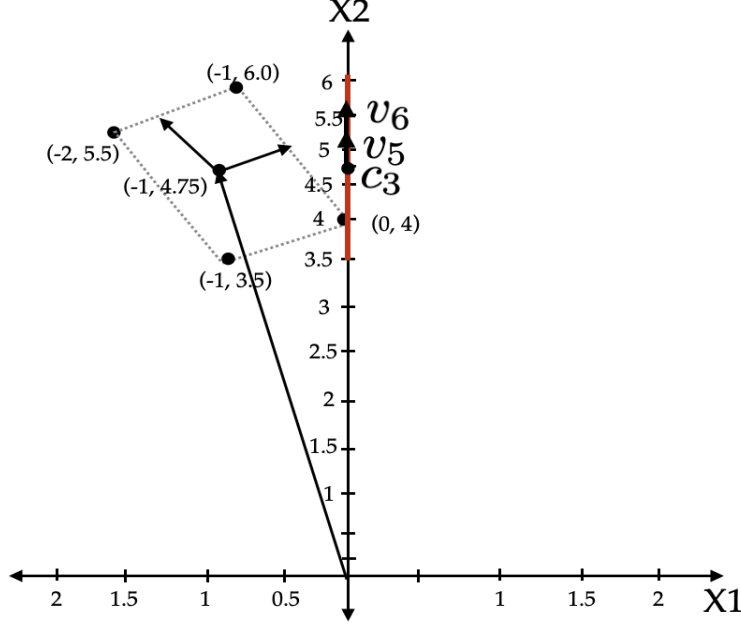


Figure 5: Star set after zeroing-out $x_1 < 0$

Applying ReLU to x_2 . We now apply ReLU to the second coordinate x_2 , separately on the two star sets obtained after processing x_1 :

$$(c_2, \{v_3, v_4\}, P_2) \quad \text{and} \quad (c_3, \{v_5, v_6\}, P_3)$$

Intersection with $x_2 \geq 0$.

From the pictures of both the star sets, it is clear that both the star sets already have $x_2 \geq 0$, so nothing will change in the sets- the sets remain as is.

Mathematical Interpretation. The second coordinate has the form

$$x_2 = 4.75 + 0.25\alpha_1 + \alpha_2.$$

Since $\alpha_1, \alpha_2 \in [-1, 1]$, we have

$$x_2 \geq 4.75 - 0.25 - 1 = 3.5 > 0.$$

Hence, intersecting with $x_2 \geq 0$ does not further restrict either star set. Both star sets remain unchanged under this branch.

Intersection with $x_2 < 0$. From the pictures of both the star sets, it is clear that both the star sets don't have any member with $x_2 < 0$, so this is empty set.

Result after ReLU at the hidden layer.

After applying ReLU to both coordinates, the non-empty star sets are:

$$(c_2, \{v_3, v_4\}, P_2) \quad \text{and} \quad (c_3, \{v_5, v_6\}, P_3)$$

Worksheet Questions

Exercise: Output layer affine transformation. We now apply the output layer affine map (W_2, b_2) to the two star sets obtained after the ReLU at the hidden layer:

$$(c_2, \{v_3, v_4\}, P_2) \quad \text{and} \quad (c_3, \{v_5, v_6\}, P_3)$$

Recall that

$$W_2 = \begin{bmatrix} 1 & -0.5 \end{bmatrix}, \quad b_2 = 1.$$

1. Apply (W_2, b_2) to the star set $(c_2, \{v_3, v_4\}, P_2)$.
 - Compute the new center $c_4 = W_2 c_2 + b_2$.
 - Compute the new basis vectors $v_7 = W_2 v_3$ and $v_8 = W_2 v_4$.
 - Write the resulting star set in the form $(c_4, \{v_7, v_8\}, P_2)$.
2. Apply (W_2, b_2) to the star set $(c_3, \{v_5, v_6\}, P_3)$.
 - Compute the new center $c_5 = W_2 c_3 + b_2$.
 - Compute the new basis vectors $v_9 = W_2 v_5$ and $v_{10} = W_2 v_6$.
 - Write the resulting star set in the form $(c_5, \{v_9, v_{10}\}, P_3)$.

Note. Since this is the output layer, no ReLU is applied after this step.

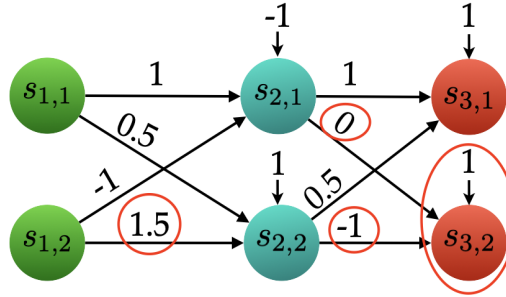


Figure 6: Updated neural network

Exercise: Star-set propagation through the updated network. Consider the updated neural network shown in Fig. 6. Start with the same input star set

$$(c_1, \{v_1, v_2\}, P_1).$$

- **Step 1: First affine layer.**

Apply the affine map (W_1, b_1) of the updated network to the input star set. Write the resulting star set in the form

$$(c_1, \{v_1, v_2\}, P_1) \xrightarrow{(W_1, b_1)} (c_2, \{v_3, v_4\}, P_1).$$

- **Step 2: ReLU on the first hidden coordinate.**

Starting from $(c_2, \{v_3, v_4\}, P_1)$, apply ReLU to the first coordinate.

- Intersect with $x_1 \geq 0$ to obtain the resulting star set.
- Intersect with $x_1 \leq 0$, apply the ReLU linear transformation, and obtain the resulting star set (if non-empty).

- **Step 3: ReLU on the second hidden coordinate.**

Apply the same two-case procedure ($x_2 \geq 0$ and $x_2 \leq 0$) to both the obtained star sets. Keep only the non-empty resulting star sets.